

SPEC Driven Development

Main Pillar of Agentic Coding

Vibe coding **V/S** Agentic Coding

- In vibe coding the biggest problem is about control. AI takes the main decision like password rules, JWT authentication etc.
- User just gives conversational prompt to AI agents.
- Here we get code fast but then end up in a loop of corrections and patches.

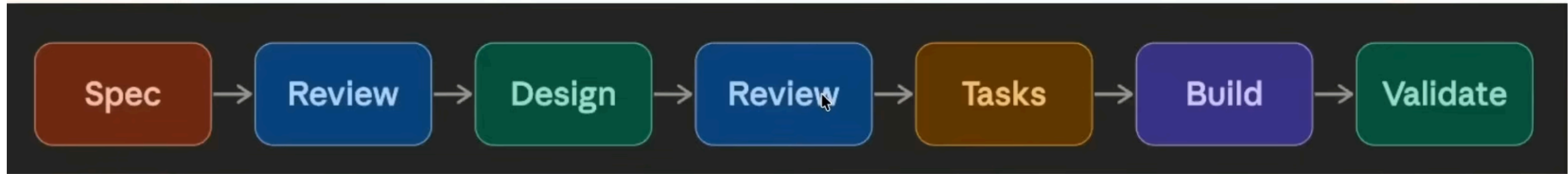
Agentic coding uses Spec driven development (SDD) this is a software development approach where very detailed specification document is written before any code is written

Spec driven development usually contains **problem statement**, **functional requirements**, **API contracts**, **constraints**, **Acceptance criteria**, **Edge cases** and **error handling**.

How is Spec Driven Development

In this we give AI agent a spec(specification) document.

Spec document is usually created by product management team and implemented by engineering team.



Spec driven development is like Developing detailed **HLD/LLD** overview and then using AI as its assistant.

Sometimes in spec driven development AI does over engineering.

Here, we get to see better debugging approach by checking against spec document unlike Vibe coding.